



IILM UNIVERSITY

Greater Noida, Uttar Pradesh

School of Computer Science and Engineering

AI Automation Internship

Lead Generation and Outreach System

STUDENT	ROLL NUMBER	SECTION	BATCH
Abhay Singh Nagarkoti	2410030450	3CSE3	2024-28

HOST ORGANIZATION

CashCow Labs | 8 June 2026 to 31 August 2026 | 12 Weeks

Candidate and Program Overview

Candidate Details

STUDENT NAME

Abhay Singh Nagarkoti

ROLL NUMBER

2410030450

SECTION

3CSE3

DEGREE

B.Tech, Computer Science and Engineering

UNIVERSITY

IILM University, Greater Noida

Internship Profile

ORGANIZATION

CashCow Labs

ROLE

AI Automation Intern

DURATION

12 Weeks, 8 June to 31 August 2026

MODE

Remote

REPORTING TO

Sharique, CEO and Co-Founder

CashCow Labs

A limited liability company registered in Wyoming, United States. It works with B2B, D2C and B2C businesses and automates the processes that normally take up the most manual hours.

Legal Structure	Limited Liability Company (LLC)
Registered In	Wyoming, United States
Domain	AI and Business Process Automation
Clients Served	B2B, D2C and B2C businesses
Mode of Internship	Remote

THE APPROACH

The company does not start by picking a tool. It maps how a careful person would do the task, step by step, and only then decides which steps can be handed to software.

Automation that skips this step tends to produce output that is fast and useless.

Clients and Collaborations

Sirion

ENTERPRISE SOFTWARE

Contract lifecycle management platform, over 7 million contracts managed

Wise

FINTECH

Cross border payments, around 19 million active customers

Ali Abdal

CREATOR ECONOMY

Creator and author, roughly 6.7 million YouTube subscribers

Greg Isenberg

VENTURE AND COMMUNITY

CEO of Late Checkout, previously advised Reddit and TikTok

BenAI

AI AUTOMATION

Agency run by Ben van Sprundel, who publishes n8n and AI agent build tutorials on YouTube

Nas Daily

MEDIA

Founded 2016, roughly 14 million YouTube subscribers

Also: SlidesAI.io, an AI presentation tool with over 14 million reported installs. The client mix is wide on purpose, so the system had to work across industries rather than for one.

Twelve Weeks, Stage by Stage

WK 1-2

Onboarding and schema

n8n fundamentals, Airtable base with five linked tables

WK 3-4

Lead sourcing

Three routes built and made switchable per profile

WK 5-6

Lead qualification

Agents returning a structured verdict with a written reason

WK 7-8

Company and competitor research

Detail scraping plus a Jina based research layer

WK 9-10

Intelligence and drafting

Pain points, solutions, hooks, then three drafts per lead

WK 11-12

Sending, follow up, hardening

Gmail sending, reply detection, rate limits and error paths

The Problem

Doing it by hand

Find people matching the profile, check the role is current, read the company, work out its problems, write the email, send it, remember to follow up, remember who already replied.

Done properly that is fifteen to twenty minutes per prospect.
Roughly twenty five prospects a day, and the research quality drops as the day goes on.

15-20 min

per prospect, and it does not scale

The usual shortcut

Give up on research and send the same email to a thousand people.
Cheap, fast, and the reason most cold email gets ignored.

Existing sales tools mostly automate the sending, which is the easiest part, and leave the research to the human.

The gap

Keep the quality of manual research, remove the manual time, and stay pointable at a new customer profile without rewriting any logic.

Project Objectives

1

Multiple sourcing channels

One provider blocking should not stop the pipeline

2

Automatic qualification

Judged against a profile stored as data, not as code

3

Deep research per lead

Company, competitors, pain points and solutions

4

Verified contact data only

Skip the lead rather than guess an address

5

Personalised drafts, not templates

Three angles generated, one selected

6

Scheduled sending and follow up

Never following up with someone who replied

A seventh objective ran through all of them: keep every intermediate result in one database so any lead can be traced from the moment it was found to the moment it was contacted.

Technologies and Tools

n8n

ORCHESTRATION

Runs the whole workflow. Triggers, branching, looping, batching and error paths. Self hosted.

Apify

LEAD SOURCING

Four actors. Cookie free LinkedIn profile search, profile detail, company details, plus a Google search scraper.

Prospero

CONTACT DATA

People search by title, headcount, industry and location. Verified email lookup by name and domain.

Jina AI

WEB RESEARCH

s.jina.ai for search and r.jina.ai for page fetch, returning clean text instead of raw HTML.

OpenAI

LANGUAGE MODELS

GPT-4.1-mini for reasoning heavy stages, GPT-4o-mini for lighter formatting work.

Airtable

DATABASE

Five linked tables holding the profile, every lead, all research, all drafts and all status flags.

Gmail

EMAIL

Sending through agent tool nodes, plus a reply search before any follow up is drafted.

LangChain nodes

AGENT FRAMEWORK

Eleven agents and four chains, with structured output parsers enforcing a JSON shape.

n8n orchestration · 179 nodes · 7 independent triggers · Airtable as the single source of truth



Seven Independent Pipelines

The system is not one long chain. Each pipeline has its own trigger, reads the records ready for it, writes the result back and sets a flag. A failure costs one stage, not the whole run.

1	Lead Sourcing and Qualification	Webhook	46 nodes
2	Company and Competitor Analysis	Every 3 hours	32 nodes
3	Profile Intelligence	Hourly	18 nodes
4	Email Enrichment	Every 6 hours	13 nodes
5	Email Drafting	Every 3 hours	11 nodes
6	Email Sending	Daily at 08:00	7 nodes
7	Follow Up	Daily at 10:00	9 nodes

Airtable as the Single Source of Truth

ICP

The customer profile definitions. Job title, decision maker, location, company size, industry and which sourcing method to use.

Lead Qualification

Every scraped lead with headline, about text, current position, profile URL, qualification status and the written reason for it.

Leads Company Details

Company description, industry, size, locations, specialities, target market, key services, tech stack and the competitor analysis.

Leads Strategy

The generated intelligence document, pain points, matched solutions and engagement hooks.

Leads Outreach

Verified email address, the three drafts, the selected email, sent status and date, follow up text and status.

Keeping state in the database rather than in the workflow makes the system restartable at any point, and makes every intermediate result visible to a person.

How it was Built

1

Map the manual process first

The design started by writing out what a careful researcher would do for one prospect, then asking which steps a machine could do at the same standard. Competitor research made the cut only because the cost dropped from hours to a few API calls.

2

Stage isolation and flag based state

Each pipeline reads records where one flag is set and its own completion flag is not. Nothing is processed twice, nothing is lost when a run stops halfway.

3

Batching and rate control

Ten batch loops and seven deliberate waits. The difference between a pipeline that finishes and one that is cut off is often a two second pause in the right place.

4

Assume the model will misbehave

Four structured output parsers enforce a JSON shape, two with auto fix. Malformed output goes back to the model with the schema attached instead of stopping the run.

RESULTS

What the System Delivers

179

nodes across seven
independent pipelines

3

sourcing channels,
switchable per profile

3

distinct email drafts
generated per lead

0

manual minutes once
a profile is configured

- Research depth per lead covers company profile, competitor analysis, an intelligence document, pain points, solutions and engagement hooks.
- No duplicate leads created, because of a two layer check inside each batch and against the database.
- No follow ups sent to prospects who had already replied, because Gmail is searched before any follow up is drafted.
- Recovery after a failed stage is automatic on the next scheduled run, with no manual restart and no reprocessing.

Challenges and Solutions

Scraping tied to a session cookie

Moved to cookie free Apify actors. No session token is stored anywhere in the workflow.

One timeout wasted the whole run

Split into seven pipelines with state tracked through completion flags in Airtable.

Models returning text that was not JSON

Structured output parsers with auto fix, returning bad output to the model with the schema.

Same profile appearing under two profiles

Duplicate check inside each batch, plus a search against Airtable by profile URL before any create.

Rate limits cutting batches off partway

Smaller batch sizes and seven wait nodes placed at the heaviest call points.

Messy company website fields

An agent step that extracts and normalises the domain before the enrichment call.

What I Took Away

Design for failure, not the happy path	The first version was one long chain. It demonstrated well and broke constantly in real runs.
State belongs in a database	Once every stage reads and writes flags on a record, the system becomes restartable and inspectable.
Prompt decomposition beats prompt length	Three focused prompts chained together produced better output than one long prompt, and were easier to debug.
Rate limits shape architecture	Batch sizes and wait placement are not tuning details added at the end. They decide whether a pipeline finishes.
Data cleaning is most of the work	Extracting a usable company domain took longer to get right than building the agent that used it.
Verified data is worth losing volume for	Skipping leads without a verified address reduces output and protects the sending domain.

Where it Goes Next

1

Reply classification

Separate interested, not interested and out of office replies, and route each one differently.

2

Multi channel sequencing

LinkedIn requests and messages alongside email, aware of each other so nobody is contacted twice in a day.

3

Feedback loop on draft selection

Record which selections led to replies, so the selection step learns from outcomes instead of judging in isolation.

4

Deliverability handling

Bounce detection, sending domain rotation and volume ramping inside the workflow.

5

Reporting layer

Reply rate by profile, by sourcing method and by email angle, so profiles can be tuned on evidence.

6

Cost tracking per lead

Compare the three sourcing routes on cost per qualified lead rather than on volume.

A seventh idea: optional human review checkpoints, where drafts wait for approval in Airtable before sending, for clients who want oversight on their first campaigns.



Internship Completion Certificate



CONCLUSION

Built to run when nobody is watching

Twelve weeks produced a working lead generation and outreach system on n8n, with 179 nodes across seven independent pipelines, integrating Apify, Prospeo, Jina AI, OpenAI, Airtable and Gmail.

The technical learning was real, but the more useful lesson was about design. The first version assumed every service would respond and every model would return valid JSON. The version that shipped assumes the opposite, and that is why it runs.

An assignment is finished when it is submitted. This system is finished when it can run for a week without anyone looking at it, and getting there took longer than building the features did.